

Designing Democratic Systems for Civic Collective Action

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Figure 1: Citizens learning about upcoming collective actions with a local civic organization (left) and interacting with paper prototypes during one of seven community deliberations from Study 1 (right). My doctoral work examines how we might design civic technologies to connect community deliberations to civic collective action in contexts like these.

ABSTRACT

Digital technologies have transformed engagement in civic collective action, connecting citizens to more action opportunities than ever before. Civic technologists have mostly focused on mobilizing grassroots collective action by providing cost-effective action opportunities and personalized ways to connect to social issues. Effective mobilization often requires citizens to deliberate about what collective action to take and how to build political power to achieve policy outcomes. Yet we know little about the deliberative processes that help citizens build motivation, develop policy solutions, and organize power to implement policy change through civic collective action. Furthermore, existing CSCW systems are not designed to help citizens learn complex information, develop the motivation to take action, build voluntary commitment, and effectively link collective action efforts to political mechanisms for implementation.

Consequently, my doctoral research aims to develop a design framework for social computing systems that empower citizens

to change local policy outcomes through deliberation in civic collective action. In this work, I will contribute a systematic design framework to civic technology researchers for supporting deliberation as the key organizing process in developing and implementing effective local policy solutions. To do so, I will provide an in-depth examination of the learning challenges in getting started with taking political action to advance theories of political participation and contribute an iterative community-based approach for developing the political will necessary to implement policy outcomes democratic deliberation.

CCS CONCEPTS

• **Human-centered computing** → **Collaborative and social computing systems and tools; Empirical studies in HCI.**

KEYWORDS

civic technology, deliberation systems, collective action, participatory policymaking

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1 OVERVIEW

Digital technologies have reshaped the ability for ordinary citizens to participate in democratic practices. From the connective power of social media in galvanizing social movements like The Arab Spring and #MeToo, to the scaling of public policy input in countries like Iceland and France afforded by crowdsourcing technologies, advances in digital technologies have opened new avenues of political participation beyond the constitutional right to vote. Tools have made organizing around common interests easier than ever [5, 9] and reduced costs and friction for mobilizing action-taking [28], giving citizens more opportunities than ever to take civic collective action: taking coordinated action to achieve a collective goal [24] while increasing their political influence by organizing power [13].

CSCW has focused on examining how the use of computing systems can effectively lower communication and coordination costs in mobilizing collective action at scale. Civic technologists in CSCW have designed mobilizing technologies that provide citizens with digital information about convenient and cost-effective opportunities to take digital action. For example, technologies like Botivist [28] and ResistBot [22] provide citizens with information about quick, easy one-time actions they can conveniently take, especially on social media. Researchers have also argued for the need to design integrated socio-technical systems for collective action focused on lowering costs associated with transitions between different stages of mobilizing collective action [e.g., 29, planning failures between coordinating and taking action].

2 OPPORTUNITIES FOR CSCW SYSTEMS

Yet, as prior scholars of digital and democratic governance have noted, increased action-taking opportunities afforded by mobilizing technologies do not necessarily translate to civic collective action that achieves policy change [1, 30]. In the aftermath of increasingly contentious election cycles and a political environment plagued by concerns of polarization, distrust in political institutions, misinformation, disinformation, and propaganda, scholars in CSCW and related fields have turned a critical eye on the role of digital technologies in exacerbating civic engagement issues at the heart of what has been called nothing short of a crisis of democracy in America today [e.g., 2, 21, 25]. Many scholars have noted problematic or even injurious consequences to the digital democratization of civic collective action, including the increase of performative online activities (e.g., liking a post, changing a profile picture) removed from aims of making policy impact [14]. Other scholars have noted that the ease and convenience of digital technologies have resulted in organizers' overemphasis on transactional mobilization goals rather than investing in leadership structures for the transformational organizing needed to make sustained policy impact [30].

We lack detailed understanding about how to design systems to support the highly complex learning and organizing challenges at the heart of helping citizens enact local, community-based policy solutions. Civic collective action that empowers citizens to participate in the process of changing the policies that govern them, not to mention developing new ones, requires more than effectively mobilizing mass participation in action-taking [e.g., 13, 15].

Consider a civic collective action effort by a community group concerned about issues of climate justice, who want their city council to pass local legislation combating climate-related inequities in their city. Here, the problem is too broadly defined for a clear policy solution to emerge, let alone one outlined with rudimentary components of a city council ordinance (e.g., regulations, stipulations, etc.) [27]. Someone or some set of citizens will have to take responsibility for developing the policy solution, which may require an unpredictable number of unpaid hours and tasks consulting policy experts, successful ordinances in other US cities, identifying and planning strategies for passing new legislation, etc. [15]. All of this is voluntary labor, which means the citizens who take on these responsibilities have to be highly motivated to do so, and usually requires that they have the financial, time, energy, intellectual, and emotional resources to dedicate (which also limits access to citizens who are more likely to have certain demographic characteristics like being older in age, more educated, and of higher socioeconomic status) [12]. Furthermore, the work itself is unpredictable and prior years of experience organizing and existing capital in their social networks will also affect a citizen's efficacy in carrying out this work.

Beyond the work of policy development, citizens also have to organize and mobilize other residents of the city to support, or at the very least not obstruct, progress on passing the proposed legislation. Even when organizing structures exist or when organizers have experience, logistical challenges remain, from the challenges of planning inclusive organizing efforts that account for variations in digital accessibility, skills, knowledge, and experience; to designing training materials; to considering the full range of tasks and projects that would benefit from having additional volunteers before aligning available volunteers with collective needs. While scholars have begun to examine some of the tasks and processes carried out by organizers in this work [e.g., 13], organizers do not follow a standard blueprint and often rely on tacit knowledge accumulated over years of practical experience or passed through their social networks to carry out these tasks.

3 PURPOSE OF DOCTORAL WORK

Thus, while civic collective action can create avenues that empower ordinary citizens to set and change political agendas, large gaps remain in our understanding of the processes underlying effective collaborating, organizing, and problem solving. How, then, can CSCW researchers design socio-technical systems that help citizens organize power and achieve local policy change through civic collective action?

The purpose of my doctoral research is to build design models for social computing systems that empower citizens to make local policy impact in civic collective action by extending democratic deliberation to community-based policymaking. Deliberation, when interdependent members of a community come together to discuss what collective action to take [31], has long been considered a foundational process in democratic decision-making that decentralizes and distributes political power among ordinary citizens [e.g., 11]. However, in practice, deliberation remains divorced from questions of power building [23], and has been criticized as idealized [21, 26] for not giving people the context-specific knowledge about where

citizens can impactfully intervene to change policy outcomes, nor the power to meaningfully set agendas or enact grassroots political change. Similarly, existing systems for supporting deliberation, like ConsiderIt [20] and Deliberatorium [18], are designed primarily to support argumentation and public reflection, treating deliberation primarily as a form of communication removed from practical considerations of power building or policy implementation. For example, while these systems have been shown to support citizens' abilities to generate and consider pros and cons for policy options, they are not designed to help the group deliberate about how the crowdsourced opinions captured on the system can be used to push legislation forward, or what information from the deliberation would be convincing to present to policymakers. As a result, we remain limited in theoretical and practical understandings of whether and how socio-technical systems can be designed to help citizens deliberate to collectively develop policy solutions and organize the political will to enact policy change.

Across the following three design research studies, my doctoral research aims to contribute a close examination of what processes and systems ought to be supported in community deliberation to reach collective action aims, from motivating public participation to iteratively updating policies and developing voluntary consent for collective actions, and how we might design CSCW systems that effectively empower communities to develop and implement policy solutions that represent their local needs.

3.1 Study 1: What problem-solving processes help communities deliberate to reach voluntary agreements in collective action, and how might socio-technical systems support these processes?

Political scientists and social computing researchers have designed deliberation processes and technologies to broaden political participation by involving community members in discussion [e.g., 16], argumentation [e.g., 18–20], and public consultation [e.g., 7]. But when communities engage in civic collective action, they often have to go beyond informed discussion to reach voluntary agreements about collective actions they agree to implement, rather than appealing to existing mechanisms (e.g., formal elections) for implementing political change. For example, when non-hierarchical grassroots organizations mobilize their members to volunteer their time to take action [4, 13], organizers often have to reach voluntary agreements about political strategies they want the group to pursue.

Across seven design iterations over eight months, I conducted a design research study [6] to examine the design of deliberation systems for reaching voluntary agreements in collective action (Lu et al., under review). Findings from Study 1 indicate that deliberation should be designed to help citizens iteratively seek information about what is preventing the group from making problem-solving progress, and key unknowns the group has to resolve before agreement can be reached. I contribute a new approach to designing deliberation systems that support iterative information seeking through well established deliberation activities like discussion and voting, as well as new processes for issue tracking and consulting

external resources. This study contributes to theories of deliberative democracy key processes of information seeking to support if deliberation aims not only to support informed discussion and decision-making, but to help deliberators iteratively make problem solving impact. For social computing research, this study aims to inform how technology can be designed to facilitate information seeking processes both during and after discussion, and identify key issue-tracking features for technologies that can aid in iterative development of political solutions that groups agree to carry out.

3.2 Study 2: How might socio-technical systems be designed to help civic organizers address challenges to taking political action that arise in the deliberation process?

Designing deliberation for helping groups reach voluntary agreements in collective action reframes the core task in deliberation from one of communication (i.e., discussion and argumentation) to one of problem solving (i.e., information seeking and iteration). When citizen deliberators are responsible for iterating on solutions before they can be voluntarily implemented, a new challenge arises: Citizens also have to take action (i.e., seeking more information, getting buy-in from additional stakeholders) following deliberation to move the collective action forward. To understand how deliberation systems can help citizens connect deliberation to action, we also have to understand how social computing systems can support organizing political action.

Building on key design implications from Study 2, this design research study [6], examines how tools for supporting reflection, planning, and social accountability, in addition to information seeking, can be leveraged in deliberation to help citizens take political action (ACM conference submission in preparation). Findings from two design iterations of a deliberation system for civic organizing indicate that citizens will have different learning needs around action-taking depending on their prior experience. These findings suggest that for most participants, organizers should take a relational and long-term approach to building trust and educating newcomers about concepts underlying political activism, as opposed to only providing them with cost-effective opportunities and logistical information. This study contributes to growing CSCW research on designing civic technologies for problem-solving impact [e.g., 3] in two ways. First, it intends to illuminate the key role of supporting relational organizing processes in motivating initial and sustained engagement with collective action efforts. Second, it seeks to demonstrate how deliberation systems can support those organizing processes by surfacing deliberation participants' motivating interests and barriers to action-taking. The intended purpose of this study is to help organizers use this information by following up with "stage-appropriate" opportunities to deepen participants' engagement in civic collective action following initial deliberation.

3.3 Study 3: What features of a deliberation system facilitate policy co-design between community members and public officials, specifically by surfacing, tracking, and responding to stakeholder needs throughout the iterative policy development process?

If a chief aim of civic collective action is to empower citizens to influence policy making beyond voting for elected representatives, or at best allowing them to vote in referendums on complete policy proposals, then a chief concern is expanding forms of public engagement throughout the policymaking process. Existing deliberation attempts, and technologies designed to support them, have framed deliberation primarily as discrete, time-bound community engagement initiatives centered around public polling or consultation, especially either very early on in the process of needfinding [e.g., 16, 17] or very late in the process to crowdsource public opinion about proposed policies [e.g., 10, 20]. These prior attempts have focused on designing to improve commentary and input from citizen stakeholders in these short deliberations, but have largely left unaccounted processes for simultaneously engaging policymakers in the iterative process of developing a publicly informed policy. Without crucially considering “the linkage problem”—how to effectively link collective action efforts among community members, particularly those supported by deliberation, with existing political mechanisms for legislative implementation [23, p. 83]—even the most ambitious and well-resourced deliberation attempts run the risk of petering out at the most crucial moments of translating deliberation outcomes into policy outcomes [e.g., 21, Ch. 7, the case of Icelandic Constitutional reform]. If deliberation is to be an effective approach to helping communities achieve civic collective action outcomes by expanding citizen policy making power, we need to better understand how deliberation can be linked to the policymaking process and support co-design between community members and public officials in doing so.

In this final proposed study, I will be taking a Design-Based Implementation Research (DBIR) approach [8] to examine the process and design of socio-technical deliberation tools for facilitating collaborative policy development between community members and local legislators. Over the course of 5 interviews with local legislators, including members of the city council and the mayor, and 10 interviews with community members, I will first develop a model of engagement outcomes and policy information that public officials desire/value from community members, and vice versa from local legislators to community members, for the task of developing new local policies on climate change. This model will inform the design of *Policy Assembly*, a tool for supporting the iterative development of a policy solution in deliberation. I will iterate on the design of *Policy Assembly* as deliberation organizers, community members and local legislators use the tool to track the development of the policy proposal over the course of 2-3 pilot community deliberations. This study aims to contribute to deliberation theory a detailed examination of how deliberation can be linked to policy making by providing a model of the engagement process and outcomes for supporting co-design between local legislators and community

members. This study will also contribute to civic technologists a design model for socio-technical tools that surface and address needs for both community members and policymakers in an iterative and deliberative process for co-designing local policies.

4 INTELLECTUAL MERIT & BROADER IMPACT

I hope to contribute a systematic design framework to civic technology researchers for supporting deliberation as the key organizing process in developing and implementing effective local policy solutions that empower communities to address their most pressing collective problems. I aim to advance theories of political participation by providing an in-depth examination of the learning challenges to getting started with taking political action, as well as contributing to theories of democratic deliberation an iterative community-based approach for developing the political will necessary to implement policy outcomes. Responding to the increasingly urgent, transdisciplinary calls for social impact in action-oriented and community-engaged research, my doctoral work aims to build a new interdisciplinary theory of deliberation in civic collective action by drawing across learning sciences, political science, and social computing.

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